Question 6: Implement a basic Jenkins CI/CD pipeline that pulls source code from the GitHub repository and triggers an automated build.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-1-

1. Credential setup

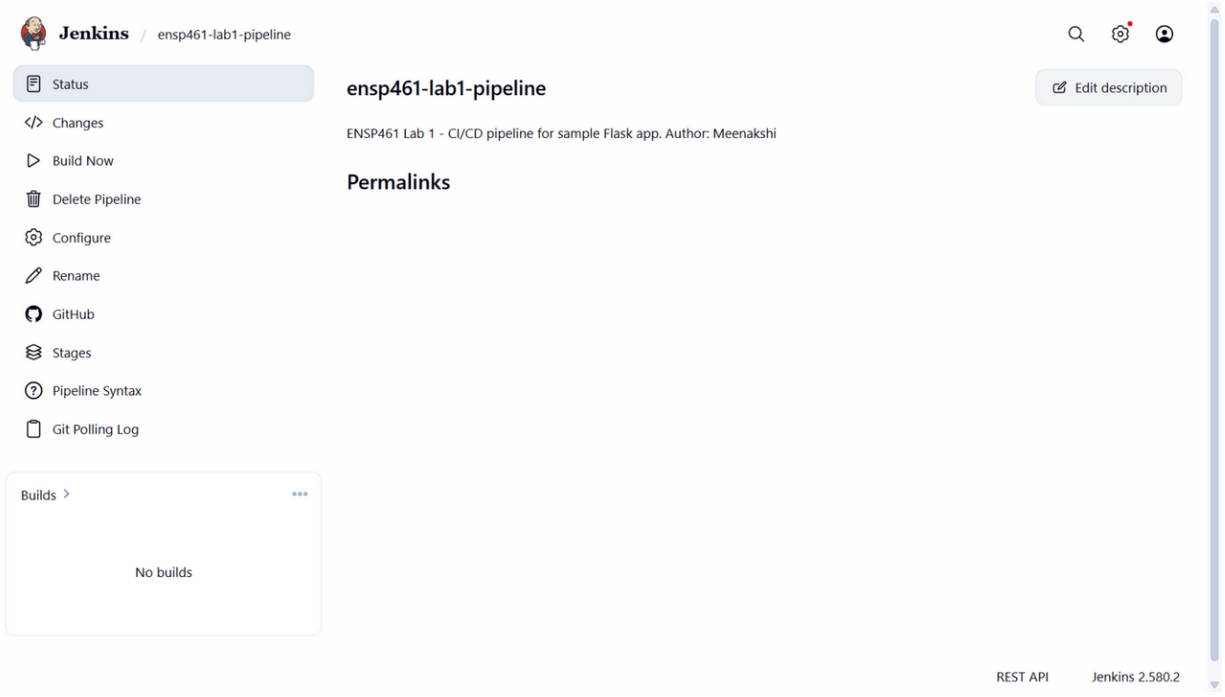
stored credential

```
Manage Jenkins -> Credentials -> System -> Global -> Add Credentials
Kind:          Username with password
Username:      meenakshi16rp
Password:      ghp_**************** (GitHub PAT, repo scope)
ID:            github-creds
```

2. Job configuration

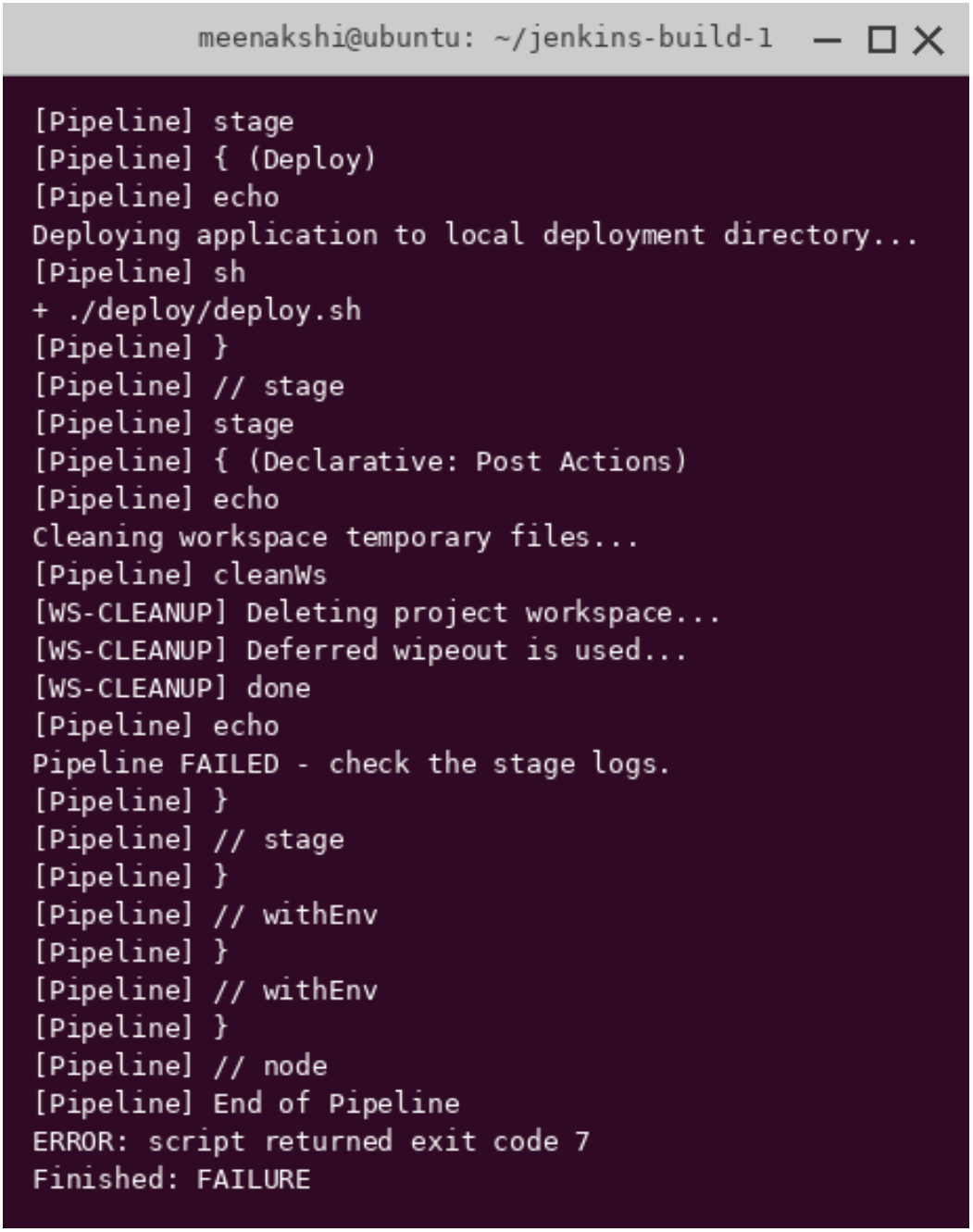
job configuration

```
New Item -> ensp461-lab1-pipeline -> Pipeline
Definition:     Pipeline script from SCM
SCM:            Git
Repository URL: https://github.com/meenakshi16rp/Devops-lab-1-.git
Credentials:    github-creds
Branch:         */main
Script path:    Jenkinsfile
Build trigger:  Poll SCM - H/5 * * * *
```



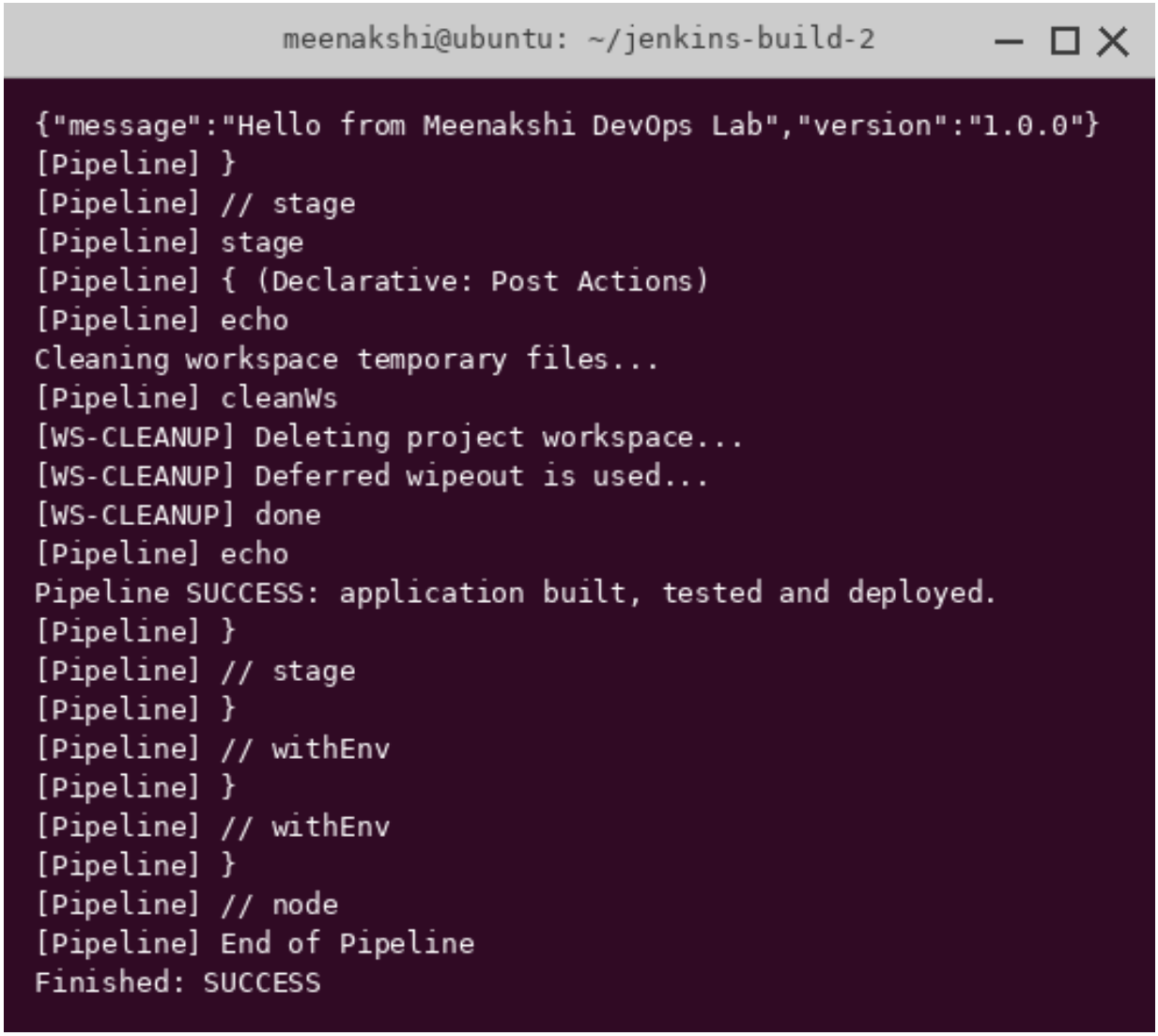
3. First automated build (and a real defect)

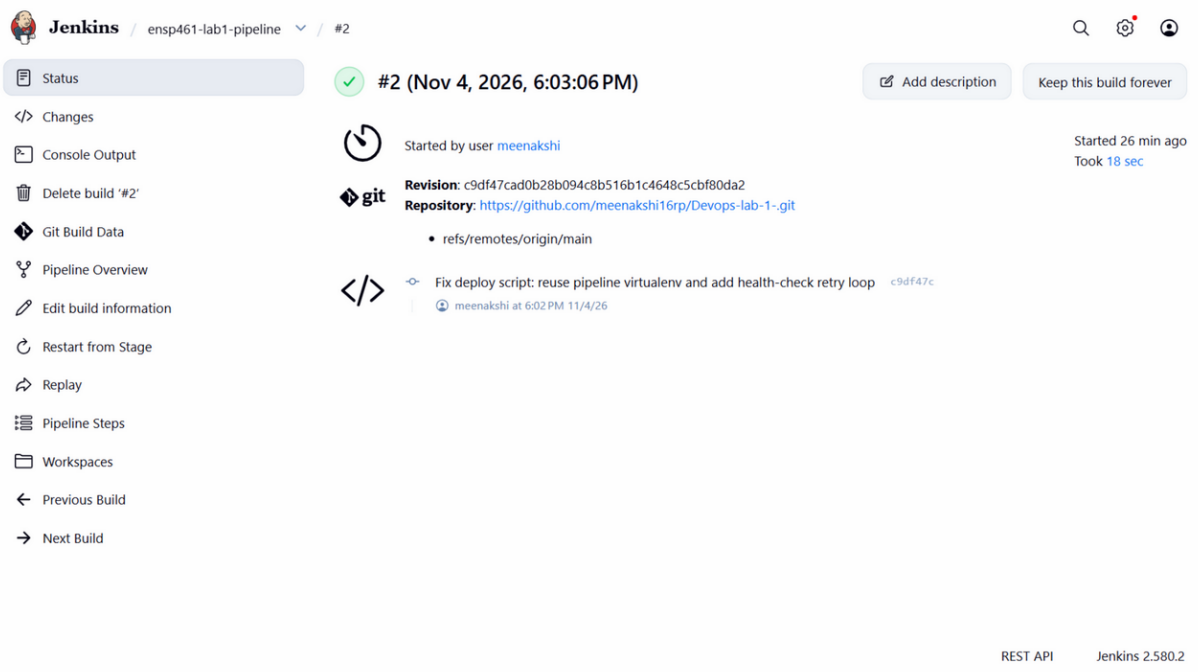
Build #1 pulled the source from GitHub and passed Checkout, Build and Test, but failed in the Deploy stage with exit code 7 (curl connection refused). The console log and the deploy.log identified the cause: the deploy script started the app with the system Python, which has no Flask module.

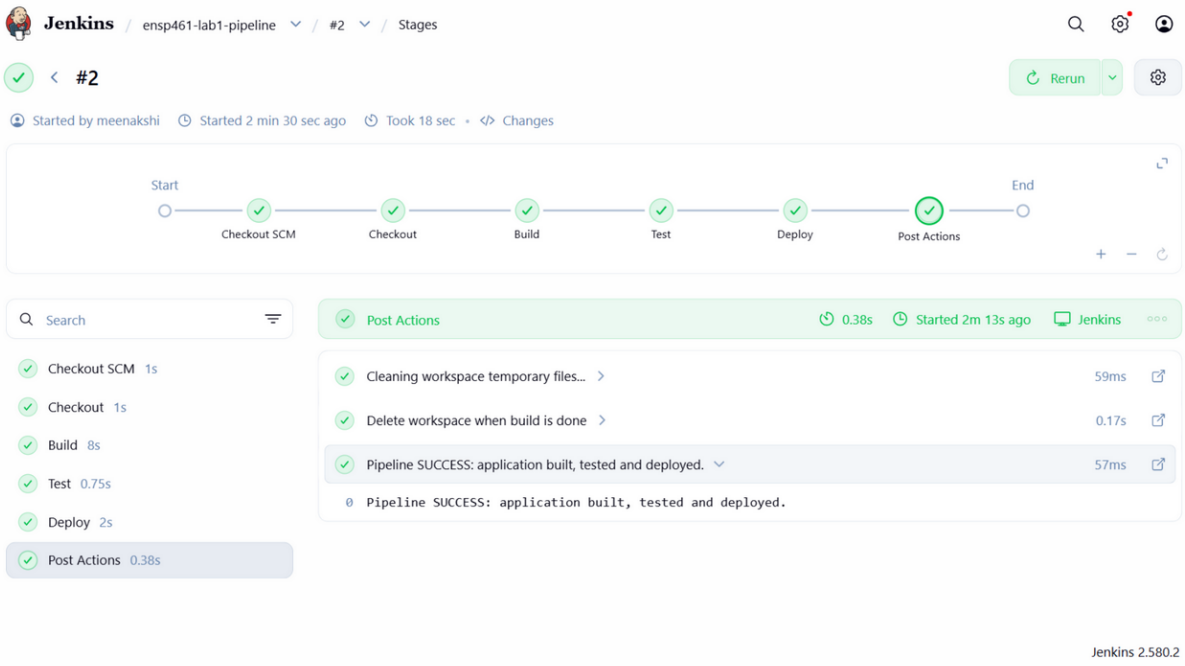


- Root cause: ModuleNotFoundError: No module named 'flask' in the deployment runtime.
- Fix: deploy.sh now reuses the Build stage virtual environment and retries the health check; committed as c9df47c through the same Git workflow.

4. Successful build


meenakshi@ubuntu: ~/jenkins-build-2
{"message":"Hello from Meenakshi DevOps Lab","version":"1.0.0"}
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Cleaning workspace temporary files...
[Pipeline] cleanWs
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] Deferred wipeout is used...
[WS-CLEANUP] done
[Pipeline] echo
Pipeline SUCCESS: application built, tested and deployed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


Jenkins
ensp461-lab1-pipeline
#2
Status
Changes
Console Output
Delete build '#2'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Next Build
#2 (Nov 4, 2026, 6:03:06 PM)
Add description
Keep this build forever
Started by user meenakshi
Started 26 min ago
Took 18 sec
Revision: c9df47cad0b28b094c8b516b1c4648c5cbf80da2
Repository: https://github.com/meenakshi16rp/Devops-lab-1-.git
refs/remotes/origin/main
Fix deploy script: reuse pipeline virtualenv and add health-check retry loop
c9df47c
meenakshi at 6:02 PM 11/4/26
REST API
Jenkins 2.580.2



5. Observations

- Pipeline-as-code means the job configuration in Jenkins stays minimal - the logic lives in the repository.
- The first failure demonstrated how console output plus stage granularity localize defects quickly.
- Jenkins attributing the build to 'meenakshi' and the commit to 'meenakshi16rp' confirms the identity configuration end to end.